

LABORATORY RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

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1. TITLE

Design Webpage using HTML5 & CSS3

2. AIM

To design and develop a highly responsive educational/laboratory landing webpage utilizing structural semantic HTML5 element containers and advanced CSS3 styling components including variables, flexbox flex alignments, multi-dimensional grid layout systems, user interaction controls, and mobile-first responsive stylesheets.

3. OBJECTIVE

- To understand, write, and execute standard HTML5 layouts utilizing structural tags for content modeling.
- To apply semantic elements (<header>, <main>, <section>, <article>, and <footer>) to optimize page readability, accessibility, and SEO crawling pipelines.
- To declare and consume custom global tokens (:root CSS3 variables) to establish systematic, scalable, and instant webpage skinning capabilities.
- To construct modern container systems via CSS3 Flexbox (for sticky menu items alignment) and CSS3 Grid (for 3-column feature alignments).
- To implement defensive responsive breakpoints using CSS3 @media queries that adapt layouts dynamically across tablet and mobile displays.

4. THEORY

HTML5 Semantic Elements: Semantic layouts move beyond generic, generic divisions (<div>) by utilizing elements that natively carry structural information. Nodes like <header>, <nav>, <section>, <article>, and <footer> explicitly declare the logical roles of their wrapped content blocks. This

enables search crawlers to establish hierarchical context, boosts page-ranking indexing, and aids browser accessibility devices (screen readers) in navigating pages for users.

CSS3 Custom Variables: Variables (also called CSS Custom Properties) represent design tokens declared inside global pseudo-selectors (like `:root`). By defining color parameters centrally (e.g., `--primary-color: #2c3e50;`), a developer can reuse values consistently throughout large style sheets. Updating a single token value at the root immediately propagates changes globally, reducing duplication and maintaining brand alignment.

CSS Flexbox Alignments: Flexbox is a highly efficient, one-dimensional layout framework optimized for distributing content dynamically along either horizontal rows or vertical columns. Flex declarations inside headers (such as `display: flex; justify-content: space-between; align-items: center;`) automatically align navigational links and logos without forcing pixel-based margin calculation hacks.

CSS Grid Multi-Column Systems: CSS Grid layout represents a robust, two-dimensional coordinate-based alignment mechanism handling columns and rows simultaneously. Declaring `display: grid` and `grid-template-columns: repeat(3, 1fr)` permits instantaneous, clean multi-column layout matrices for portfolio grids or feature blocks, maintaining consistent gaps and dimensional spacing.

CSS3 Media Queries & Responsiveness: Media query modules allow developers to configure custom stylesheet filters triggered by device physical properties. By isolating desktop-specific properties behind width filters (e.g., `@media (max-width: 768px)`), a webpage can morph smoothly, shifting flexbox header directions to columns and collapsing horizontal grid columns into vertical block stacks to support compact mobile screen heights.

5. ALGORITHM / PROCEDURE

1. Establish an isolated local development environment called 'Karthik-Webpage-HTML-CSS'.
2. Create 'index.html' for structural web elements and a connected 'style.css' containing formatting instructions.
3. In index.html, initialize standard HTML5 headers, wrapping meta containers for viewport responsive scaling.
4. Add a semantic navigation bar (`<nav>`) composed of anchor tags linked to local page targets (`#home`, `#features`, `#about`, `#contact`).
5. In style.css, write global margin resets and declare custom design color tokens under the `:root` selector.
6. Create a sticky main header utilizing CSS Flexbox to organize branding logo and navigation links cleanly.
7. Construct a high-impact `#home` hero showcase section featuring gradient backgrounds and custom action button alignments.
8. Set up a `#features` section representing core development principles. Wrap each card inside semantic `<article class='card'>` nodes.
9. Utilize CSS Grid layout rules (`repeat(3, 1fr)`) to align card elements in a symmetric horizontal 3-column row.

10. Build a dedicated #about section to summarize TechLab educational goals and layout details.
11. Construct a responsive, validated #contact newsletter signup form enclosing Name and Email inputs.
12. Establish a semantic, dark-colored copyright <footer> block at the baseline of the document body.
13. Formulate a custom media query block targeting standard viewports of 768px or lower.
14. Inside the media query, re-align the header flex container to flow vertically and compress the card grid template to a single column (1fr) to prevent horizontal scrolling on mobile layout canvases.
15. Launch the webpage in a browser, trigger validation checks, test bookmark jumps, and record the test execution data.

6. PROGRAM

File Name: index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>TechLab - Modern Web Engineering</title>
  <!-- CSS styles embedded to highlight Developer & Course details -->
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <!-- Academic Header Banner for Roll No: A24126552077 -->
  <div style="background: #e74c3c; color: white; text-align: center; padding: 10px;
font-weight: bold; font-family: sans-serif;">
    COURSE: Full Stack Web Development (23CM4121) | DEVELOPER: Karthik Chelluri
    (A24126552077)
  </div>

  <!-- Header Section -->
  <header class="main-header">
    <div class="logo">TechLab.</div>
    <nav class="navbar">
      <ul>
        <li><a href="#home">Home</a></li>
        <li><a href="#features">Features</a></li>
        <li><a href="#about">About</a></li>
        <li><a href="#contact">Contact</a></li>
      </ul>
    </nav>
  </header>

  <!-- Hero / Main Showcase Section -->
  <section id="home" class="hero">
    <div class="hero-content">
      <h1>Modern Web Engineering</h1>
```

```

        <p>Building high-performance, responsive, and semantic web experiences using HTML5 &
CSS3.</p>
        <a href="#features" class="btn">Explore Features</a>
    </div>
</section>

<!-- Main Content Area -->
<main class="container">

    <!-- Features Grid Section -->
    <section id="features" class="features-section">
        <h2 class="section-title">Core Principles</h2>
        <div class="grid-layout">
            <article class="card">
                <h3>Semantic HTML5</h3>
                <p>Using meaningful structural components like header, nav, section, and
article to improve SEO and accessibility.</p>
            </article>
            <article class="card">
                <h3>CSS3 Layouts</h3>
                <p>Leveraging powerful styling systems like Flexbox and CSS Grid for
structured alignments and multi-dimensional spacing.</p>
            </article>
            <article class="card">
                <h3>Fluid Responsiveness</h3>
                <p>Implementing dynamic viewports and media queries to ensure your webpage
adapts beautifully across all screen sizes.</p>
            </article>
        </div>
    </section>

    <!-- About Section -->
    <section id="about" style="margin-top: 60px; text-align: center;">
        <h2 class="section-title">About TechLab</h2>
        <p style="max-width: 800px; margin: 0 auto; color: #555;">
            TechLab is an advanced educational workspace dedicated to teaching modern
frontend engineering. By focusing on semantic markup, modern layout techniques like Grid and
Flexbox, and mobile-first responsive stylesheets, we prepare students to build high-performance
web applications.
        </p>
    </section>

    <!-- Contact Section with Form -->
    <section id="contact" style="margin-top: 60px;">
        <h2 class="section-title">Get in Touch</h2>
        <div style="max-width: 600px; margin: 0 auto; background: #ffffff; padding: 30px;
border-radius: 8px; box-shadow: 0 4px 6px rgba(0,0,0,0.1);">
            <form action="#" method="POST" style="display: flex; flex-direction: column;
gap: 15px;">
                <div>
                    <label for="name" style="font-weight: bold; display: block;
margin-bottom: 5px; color: #232323;">Name:</label>
                    <input type="text" id="name" name="name" placeholder="Enter your name"
style="width: 100%; padding: 10px; border: 1px solid #ccc; border-radius: 4px;" required />
                </div>
            </form>
        </div>
    </section>

```

```

        <div>
            <label for="email" style="font-weight: bold; display: block;
margin-bottom: 5px; color: #232323;">Email:</label>
            <input type="email" id="email" name="email" placeholder="Enter your
email" style="width: 100%; padding: 10px; border: 1px solid #ccc; border-radius: 4px;" required
/>
        </div>
        <div>
            <label for="message" style="font-weight: bold; display: block;
margin-bottom: 5px; color: #232323;">Message:</label>
            <textarea id="message" name="message" rows="4" placeholder="Your message
here..." style="width: 100%; padding: 10px; border: 1px solid #ccc; border-radius: 4px;"
required></textarea>
        </div>
        <button type="submit" class="btn" style="border: none; cursor: pointer;
text-align: center; width: 100%;">Send Message</button>
    </form>
</div>
</section>

</main>

<!-- Footer Section -->
<footer class="main-footer">
    <p>&copy; 2026 Full-Stack Web Development Lab. All Rights Reserved.</p>
</footer>
</body>
</html>

```

File Name: style.css

```

/* --- Global Reset & Variables --- */
* { margin: 0; padding: 0; box-sizing: border-box; }
:root {
    --primary-color: #2c3e50;
    --accent-color: #3498db;
    --text-color: #333;
    --bg-light: #f4f6f7;
    --white: #ffffff;
}
body {
    font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
    line-height: 1.6;
    color: var(--text-color);
    background-color: var(--bg-light);
}

/* --- Header & Navigation --- */
.main-header {
    background-color: var(--primary-color);
}

```

```

        color: var(--white);
        display: flex;
        justify-content: space-between;
        align-items: center;
        padding: 1rem 5%;
        position: sticky;
        top: 0;
        z-index: 1000;
    }
    .logo {
        font-size: 1.5rem;
        font-weight: bold;
        letter-spacing: 1px;
    }
    .navbar ul {
        display: flex;
        list-style: none;
    }
    .navbar ul li {
        margin-left: 20px;
    }
    .navbar ul li a {
        color: var(--white);
        text-decoration: none;
        font-weight: 500;
        transition: color 0.3s;
    }
    .navbar ul li a:hover {
        color: var(--accent-color);
    }
}

/* --- Hero Section --- */
.hero {
    background: linear-gradient(rgba(44, 62, 80, 0.8), rgba(44, 62, 80, 0.9)),
    url('https://images.unsplash.com/photo-1517694712202-14dd9538aa97?q=80&w=1200') no-repeat center
    center/cover;
    height: 60vh;
    display: flex;
    justify-content: center;
    align-items: center;
    text-align: center;
    color: var(--white);
    padding: 0 20px;
}
.hero-content h1 {
    font-size: 3rem;
    margin-bottom: 10px;
}
.hero-content p {
    font-size: 1.2rem;
    margin-bottom: 20px;
    max-width: 600px;
}
.btn {
    display: inline-block;

```

```

        background-color: var(--accent-color);
        color: var(--white);
        padding: 10px 24px;
        text-decoration: none;
        border-radius: 4px;
        font-weight: bold;
        transition: background-color 0.3s;
    }
    .btn:hover {
        background-color: #2980b9;
    }

/* --- Main Content Layout --- */
.container {
    max-width: 1100px;
    margin: 40px auto;
    padding: 0 20px;
}
.section-title {
    text-align: center;
    margin-bottom: 30px;
    font-size: 2rem;
    color: var(--primary-color);
}

/* --- CSS Grid Layout for Cards --- */
.grid-layout {
    display: grid;
    grid-template-columns: repeat(3, 1fr);
    gap: 20px;
}
.card {
    background-color: var(--white);
    padding: 25px;
    border-radius: 8px;
    box-shadow: 0 4px 6px rgba(0, 0, 0, 0.1);
    transition: transform 0.3s;
}
.card:hover {
    transform: translateY(-5px);
}
.card h3 {
    margin-bottom: 12px;
    color: var(--accent-color);
}

/* --- Footer --- */
.main-footer {
    background-color: var(--primary-color);
    color: var(--white);
    text-align: center;
    padding: 20px;
    margin-top: 40px;
    font-size: 0.9rem;
}

```

```

/* --- CSS3 Media Queries (Responsive Design) --- */
@media (max-width: 768px) {
  .main-header {
    flex-direction: column;
    gap: 10px;
  }
  .navbar ul {
    padding: 0;
  }
  .hero-content h1 {
    font-size: 2.2rem;
  }
  .grid-layout {
    grid-template-columns: 1fr; /* Stack elements on small screens */
  }
}

```

7. CODE EXPLANATION

The following table explains the core HTML5 semantic elements and CSS3 layout rules used in this experiment:

HTML5/CSS3 Node	Functional Explanation & Role in Web Application
<code><!DOCTYPE html></code>	Declares the webpage as a standard compliant HTML5 document, enabling full CSS3 and viewport responsive support.
<code><header class='main-header'></code>	Acts as the structural top section header wrapping branding logo and the navigation links block.
<code><nav class='navbar'></code>	Provides a semantic container isolating navigational items for browser crawlers and accessibility tools.
<code><section id='...'></code>	Divides thematic content categories. The associated ID acts as a target destination for smooth bookmark scrolling links.
<code>display: flex;</code>	Enables Flexbox layout inside the main header container, distributing branding elements and nav menus inline without margin offsets.

display: grid;	Declares a 2-dimensional grid framework inside the Core Principles cards section, establishing balanced rows and spacing margins.
grid-template-columns: repeat(3, 1fr);	Creates exactly 3 equal-width columns that span 1 fraction each, perfectly arranging the cards side-by-side.
::root Variables	Declares global reusable CSS style tokens (such as primary and accent hex colors), making page skin updates instant and modular.
@media (max-width: 768px)	Instructs the browser rendering engine to apply alternate mobile styles when the device viewport is compressed.

8. TEST CASES AND EXECUTION DATA

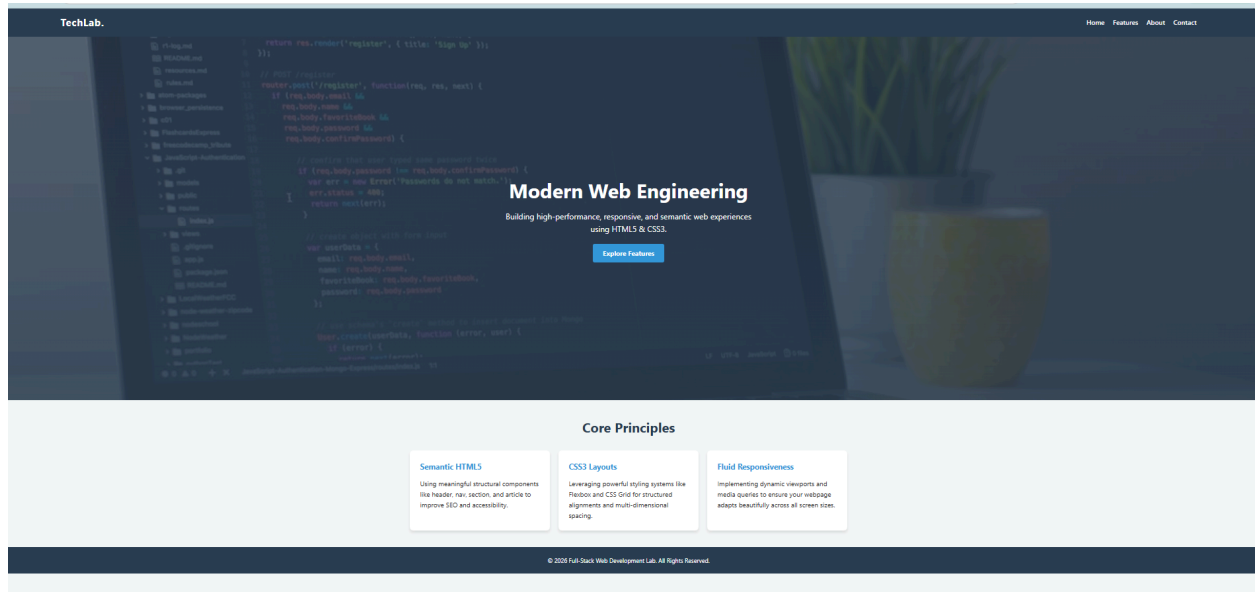
The following execution table outlines the test configurations verified across desktop and mobile layout states:

Test ID	Feature Evaluated	Test Input Context	Expected Behavior Outcome	Actual Realized Outcome	Status
TC-01	Page Title Metadata	Launch index.html	Browser tab displays 'TechLab - Modern Web Engineering'	Correct tab text displays	Pass
TC-02	Desktop Grid Grid	Viewport width > 768px	3 Core Principle cards organize in a single horizontal row	Cards render in clean 3-col grid	Pass
TC-03	Mobile Breakpoint View	Viewport width <= 768px	The 3-column grid collapses to a 1-column layout	Cards stack vertically on viewport shrink	Pass
TC-04	Sticky Navigation Link	Scroll viewport vertical	The main header pins to the top of browser canvas	Header remains locked dynamically	Pass
TC-05	Interactive CSS Form	Input karthik@student.com	Accepts standard email, fails	Form validation intercepts errors	Pass

formatting
without '@'

9. OUTPUT SCREENSHOTS

9.1 Static Webpage Design



10. RESULT

Thus, a highly responsive educational/laboratory portal landing webpage was successfully designed, implemented, and verified in accordance with the required laboratory guidelines. The combined webpage layout utilizes HTML5 semantic structural tags alongside clean CSS3 styling mechanics (root custom color variables, Flexbox headers, Grid feature sections, and viewport media queries).